<u>מבוא למדעי כדור הארץ – סיסמיקה וכבידה, תרגיל 4</u>

קראו תת-פרק 5.5.2 (מהספר של Fowler) וענו על השאלות הבאות.

מתוך הספר של Turcotte and Schubert :

Problem 5.23 Consider the formation of a sedimentary basin on the seafloor. Suppose isostatic compensation is achieved by the displacement of mantle material of density ρ_m. Show that sediment thickness s is related to water depth d by

$$s = \frac{(\rho_m - \rho_w)}{(\rho_m - \rho_s)}(D - d), \tag{5.150}$$

where D is the initial depth of the sediment-free ocean. What is the maximum possible thickness of the sediment if $\rho_s = 2500$ kg m^{-3}, $\rho_m = 3300$ kg m^{-3}, and $D = 5$ km?

מתוך הספר של Fowler :

13. Calculate the depths and densities beneath a 5-km-high mountain chain in isostatic equilibrium with a 35-km-thick continental crust of density 2.8×10^3 kg m^{-3} and a mantle of density 3.3×10^3 kg m^{-3} by using the hypotheses of (a) Pratt and (b) Airy.
14. A mountain range 4 km high is in isostatic equilibrium.
    (a) During a period of erosion, a 2 km thickness of material is removed from the mountains. When the new isostatic equilibrium is achieved, how high are the mountains?
    (b) How high would they be if 10 km of material were eroded away?
    (c) How much material must be eroded to bring the mountains down to sea level? (Use crustal and mantle densities of 2.8×10^3 kg m^{-3} and 3.3×10^3 kg m^{-3}, respectively.)